



Spring AOP and REST

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6. Implementation

- Some logic needs to be called across multiple points of a web application.
- This is called cross cutting concern and referred as aspect in spring AOP
- It helps to separate main business logic from extra tasks which are required at multiple places and adds modularity to the application
- These extra tasks can be configured easily in the application
- Example of such cross cutting concerns are logging, security, auditing performance, compressing response etc.
- Implementation without AOP
repeated code for doing the same thing at multiple places
- Implementation with AOP
repeated code at single place and configured in XML

2. Terminologies and advice types

- Joinpoint
 - A location in the application where cross cutting concern needs to be executed. Spring AOP supports only method execution join points.
- Advice
 - Action taken by an aspect at a particular joinpoint. There are 5 types of advice in Spring AOP.
 1. Before Advice
 2. After returning advice
 3. After throwing advice
 4. After advice
 5. Around advice
- Pointcut
 - Expression language that matches join points.
 - Advice needs to be associated with a pointcut expression

- Proxy.
 - It is an object that is created after applying advice to the target object
- Target object
 - It is the object i.e. being advised by one or more aspects
- Aspect
 - It is a class that contains advices, joinpoints etc.
- Weaving
 - It is the process of linking aspect with objects to create an advised object. Weaving can be done at compile time, load time or runtime. Spring AOP performs weaving at runtime.

- Add maven dependencies in the pom.xml file for spring-aop and aspectjweaver.
- Creating target class having business logic
- Declaring advice with pointcut declaration in the class annotated with `@Aspect`. Advice is a method annotated with it's specific type of annotation and empty methods annotated by `@Pointcut` annotation for pointcut declaration
- Adding configuration in bean definition file
- Testing the advices by calling business logic methods

- What is web service
 - A web service is a software module that performs set of tasks.
 - It is hosted on the server
 - Client execute web service calls(requests) to the server hosting the web service
- Need of web service
 - Web services provide a common platform that allows multiple applications built on various programming languages to have the ability to communicate with each other.
- Types of Web Services
 1. SOAP web services.
 2. RESTful web services.

- REST Web service
 - REST stands for REpresentational State Transfer.
 - The underlying protocol for REST is HTTP.
 - The calling client can perform predefined operations.
 - The information(representation) is delivered in several formats mainly JSON.
 - In short it is service running on web server accessed via HTTP and generated response is majorly in the form of JSON as it is language independent.

5. REST methods

HTTP Method	Operation	Comment
GET	Read Operation only	Uses only for the read operation.GET should be idempotent
POST	Create new resource	Should only be used to create a new resource
PUT	Update / Replace Resource	Update an existing resource. Think of PUT method as modifying a resource
DELETE	Delete Resource	To remove a given resource.DELETE operation is idempotent
PATCH	Partial Update / Modify	Partial update to a resource should happen through PATCH

- Spring MVC supports REST from version 3.0
- Easy to develop because of built in annotations
- How request flows
 - All HTTP Requests are intercepted by DispatcherServlet
 - Handler Mappings help to find controllers
 - Requests are now processed by the Controller
 - Data can be directly returned from Controller
- RestController annotation
 - This annotation is actually combination of @Controller and @ResponseBody.
 - @RequestMapping is used for mapping request uri to particular method in the controller.